

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM-7

14-07-2026

7. Write the python program to implement BFS.

Aim

To write a Python program to implement the Breadth-First Search (BFS) algorithm for graph traversal using user input.

Algorithm

1. Start the program.
2. Define the graph using an adjacency list.
3. Select the starting vertex.
4. Create an empty queue and a set to store visited vertices.
5. Insert the starting vertex into the queue and mark it as visited.
6. Remove the first vertex from the queue and add it to the traversal list.
7. Examine all adjacent vertices of the selected vertex.
8. Add each unvisited adjacent vertex to the queue and mark it as visited.
9. Repeat the process until the queue becomes empty.
10. Display the BFS traversal order.
11. Use NetworkX and Matplotlib to generate the graph and display the visiting order.
12. Stop the program.

Program

```
import networkx as nx
```

```
import matplotlib.pyplot as plt
```

```
from collections import deque
```

```
g = {'A':['B','C'],'B':['D','E'],'C':['F'],'D':[],'E':[],'F':[]}
```

```
s = 'A'
```

```
q = deque([s])
```

```

v = {s}
order = []

while q:
    n = q.popleft()
    order.append(n)
    for x in g[n]:
        if x not in v:
            v.add(x)
            q.append(x)

print("BFS:", " -> ".join(order))

G = nx.Graph()
for n in g:
    for x in g[n]:
        G.add_edge(n,x)

pos = {'A':(0,2),'B':(-1,1),'C':(1,1),
       'D':(-1.5,0),'E':(-.5,0),'F':(1,0)}

nx.draw(G,pos,with_labels=True,node_size=1800,
        node_color="lightblue")

for i,n in enumerate(order):
    plt.text(pos[n][0],pos[n][1]+.2,str(i+1),
             fontsize=14)

plt.title("Breadth First Search (BFS)")
plt.show()

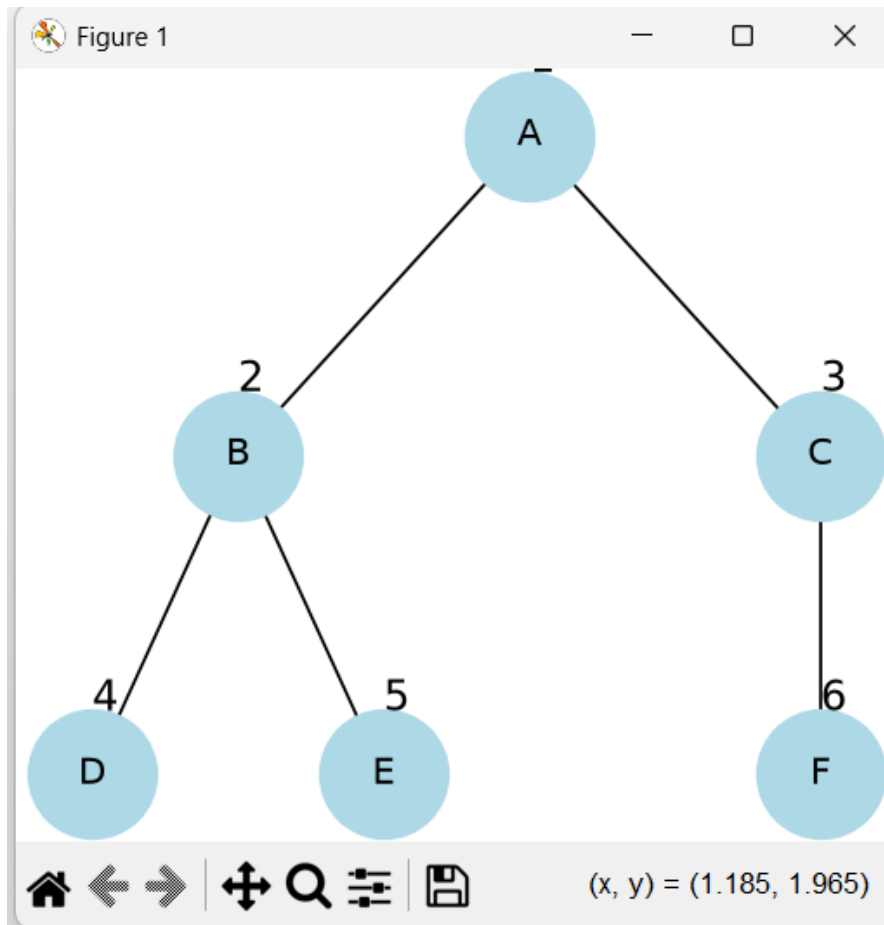
```

Input

Starting node: A

Output

BFS: A -> B -> C -> D -> E -> F



Result:

Thus, the Python program to implement the Breadth-First Search (BFS) algorithm using user input was executed successfully, and the graph was traversed in BFS order with the desired output obtained.